

Module 16

4. CSS Selectors & Styling

Theory Assignment

Question 1: What is a CSS selector? Provide examples of element, class, and ID selectors.

A CSS selector is used to select HTML elements for styling.

Types of Selectors

Element Selector

Selects HTML elements directly.

```
p{  
    color: blue;  
}
```

Class Selector

Selects elements using class name.

```
.box{  
    background-color: lightgray;  
}
```

ID Selector

Selects a unique element using ID.

```
#header{  
    color: red;  
}
```

Question 2: Explain the concept of CSS specificity. How do conflicts between multiple styles get resolved?

CSS specificity decides which style will be applied when multiple styles target the same element.

Priority Order

1. Inline CSS
2. ID Selector
3. Class Selector
4. Element Selector

The style with higher specificity overrides lower specificity styles.

Question 3: What is the difference between internal, external, and inline CSS? Discuss the advantages and disadvantages of each approach.

Inline CSS

CSS written directly inside HTML elements.

Advantage:

- Quick styling.

Disadvantage:

- Difficult to manage large projects.
-

Internal CSS

CSS written inside the `<style>` tag in the HTML file.

Advantage:

- Good for single-page websites.

Disadvantage:

- Cannot reuse styles across multiple pages.
-

External CSS

CSS written in a separate `.css` file.

Advantage:

- Reusable and easy to maintain.

Disadvantage:

- Requires additional file linking.

Lab Assignment – External CSS Contact Form

HTML File

```
<!DOCTYPE html>
<html>

<head>
    <title>Contact Form</title>

    <link rel="stylesheet" href="style.css">
</head>

<body>

    <form id="contact-form">

        <h1>Contact Form</h1>

        <label for="fullname">Full Name</label>
        <input type="text" id="fullname" class="input-field">

        <label for="email">Email</label>
        <input type="email" id="email" class="input-field">

        <button id="submit-btn">Submit</button>

    </form>

</body>

</html>
```

CSS File

```
body{
    background-color: lightgray;
```

```
}

#contact-form{
  background-color: white;
  width: 300px;
  padding: 20px;
  margin: auto;
}

.input-field{
  width: 100%;
  padding: 10px;
  margin-bottom: 15px;
}

#submit-btn{
  background-color: blue;
  color: white;
  padding: 10px;
  border: none;
}

#submit-btn:hover{
  background-color: darkblue;
}
```

5. CSS Box Model

Theory Assignment

Question 1: Explain the CSS box model and its components (content, padding, border, margin).

The CSS box model describes how elements are displayed on a webpage.

Components

- **Content** – Actual text or image.
- **Padding** – Space inside the border.
- **Border** – Surrounds padding and content.
- **Margin** – Space outside the border.

Each component affects the total size of the element.

Question 2: What is the difference between border-box and content-box box-sizing in CSS? Which is the default?

content-box

- Width includes only content.
- Padding and border increase total size.
- Default box-sizing.

border-box

- Width includes content, padding, and border.
- Easier for responsive layouts.

Lab Assignment – Profile Card

HTML File

```
<!DOCTYPE html>
<html>

<head>
  <title>Profile Card</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <div class="card">

    

    <h2>Arbaz Siddiqui</h2>

    <p>Web Developer and Designer</p>

    <button>Follow</button>

  </div>

</body>

</html>
```

CSS File

```
body{
  display: flex;
  justify-content: center;
```

```
}

.card{
  width: 300px;
  padding: 20px;
  border: 2px solid gray;
  margin: 50px;
  text-align: center;
  box-sizing: border-box;
}

img{
  width: 100%;
  box-sizing: content-box;
}

button{
  padding: 10px 20px;
}
```

6. CSS Flexbox

Theory Assignment

Question 1: What is CSS Flexbox, and how is it useful for layout design?

CSS Flexbox is a layout model used to align and distribute elements efficiently.

Terms

- **Flex Container** – Parent element using `display:flex`.
- **Flex Item** – Child elements inside the flex container.

Flexbox helps create responsive layouts easily.

Question 2: Describe the properties `justify-content`, `align-items`, and `flex-direction` used in Flexbox.

`justify-content`

Aligns items horizontally.

`align-items`

Aligns items vertically.

flex-direction

Defines item direction such as row or column.

Lab Assignment – Flexbox Layout

HTML File

```
<!DOCTYPE html>
<html>

<head>
  <title>Flexbox Layout</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <header>Header</header>

  <div class="container">

    <aside>Sidebar</aside>

    <main>Main Content</main>

  </div>

  <footer>Footer</footer>

</body>

</html>
```

CSS File

```
.container{
  display: flex;
  justify-content: space-between;
  align-items: center;
}

aside{
  width: 20%;
  background-color: lightgray;
```

```
}

main{
  width: 75%;
}

@media (max-width:768px) {

  .container{
    flex-direction: column;
  }

}
```

7. CSS Grid

Theory Assignment

Question 1: Explain CSS Grid and how it differs from Flexbox. When would you use Grid over Flexbox?

CSS Grid is a two-dimensional layout system that controls rows and columns together.

Difference

- Flexbox works in one direction.
- Grid works in rows and columns together.

Grid is best for complex webpage layouts.

Question 2: Describe the grid-template-columns, grid-template-rows, and grid-gap properties.

grid-template-columns

Defines column sizes.

grid-template-rows

Defines row sizes.

grid-gap

Adds spacing between grid items.

Lab Assignment – 3x3 Product Grid

HTML File

```
<!DOCTYPE html>
<html>

<head>
  <title>CSS Grid</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <div class="grid-container">

    <div class="card">
      
      <h3>Laptop</h3>
      <p>₹50,000</p>
    </div>

    <div class="card">
      
      <h3>Phone</h3>
      <p>₹20,000</p>
    </div>

    <div class="card">
      
      <h3>Headphones</h3>
      <p>₹2,000</p>
    </div>

  </div>

</body>

</html>
```

CSS File

```
.grid-container{
  display: grid;
  grid-template-columns: auto auto auto;
  grid-gap: 20px;
}
```

```
.card{
  border: 1px solid gray;
  padding: 20px;
  text-align: center;
}

.card:hover{
  transform: scale(1.05);
}
```

8. Responsive Web Design with Media Queries

Theory Assignment

Question 1: What are media queries in CSS, and why are they important for responsive design?

Media queries are CSS rules used to apply styles based on screen size and device type. They help make websites responsive for mobile, tablet, and desktop screens.

Question 2: Write a basic media query that adjusts the font size of a webpage for screens smaller than 600px.

```
@media (max-width:600px) {

  body{
    font-size:14px;
  }

}
```

Lab Assignment – Responsive Webpage

HTML File

```
<!DOCTYPE html>
<html>

<head>
  <title>Responsive Layout</title>
```

```
<link rel="stylesheet" href="style.css">
</head>

<body>

  <nav>Navbar</nav>

  <div class="content">

    <div class="left">Column 1</div>

    <div class="right">Column 2</div>

  </div>

  <footer>Footer</footer>

</body>

</html>
```

CSS File

```
.content{
  display:flex;
}

.left,.right{
  width:50%;
  padding:20px;
}

@media (max-width:768px) {

  .content{
    flex-direction:column;
  }

  .left,.right{
    width:100%;
    font-size:14px;
  }

}
```

9. Typography and Web Fonts

Theory Assignment

Question 1: Explain the difference between web-safe fonts and custom web fonts.

Web-Safe Fonts

Fonts already installed on most systems.

Custom Web Fonts

Fonts downloaded from external sources like Google Fonts.

Web-safe fonts load faster and improve compatibility.

Question 2: What is the font-family property in CSS? How do you apply a custom Google Font to a webpage?

The `font-family` property defines the font style for text.

Google Fonts can be added using a link inside the head tag.

Lab Assignment – Blog Layout

HTML File

```
<!DOCTYPE html>
<html>

<head>

    <title>Blog Layout</title>

    <link rel="preconnect" href="https://fonts.googleapis.com">

    <link
href="https://fonts.googleapis.com/css2?family=Poppins&family=Roboto&display=
swap" rel="stylesheet">

    <link rel="stylesheet" href="style.css">

</head>

<body>

    <div class="blog">

        <h1>Blog Title</h1>

        <h3>Blog Subtitle</h3>
```

```
        <p>
          This is the blog content section.
        </p>
      </div>
</body>
</html>
```

CSS File

```
h1,h3{
  font-family:'Poppins',sans-serif;
}

p{
  font-family:'Roboto',sans-serif;
  line-height:1.8;
  font-size:18px;
}

.blog{
  padding:20px;
}
```